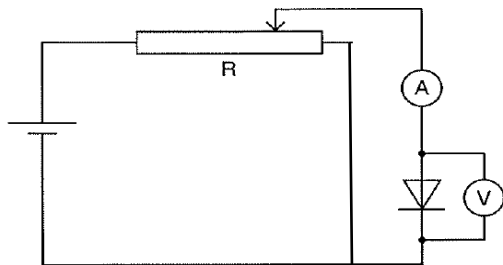
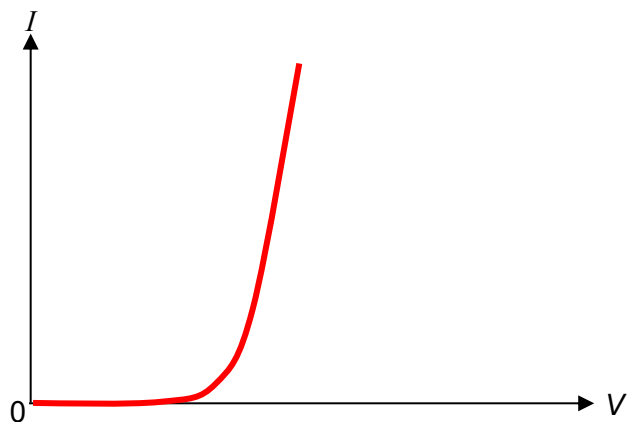


in the primary circuit.

6ai



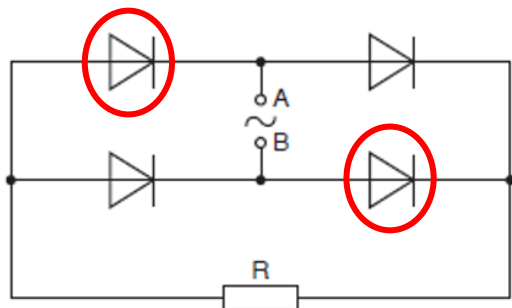
aii



aiii

The resistance can be obtained by taking the ratio of V to I

bi



ii

$$V_{\max} = \sqrt{2} \times 7.0 = 9.9 \text{ V}$$

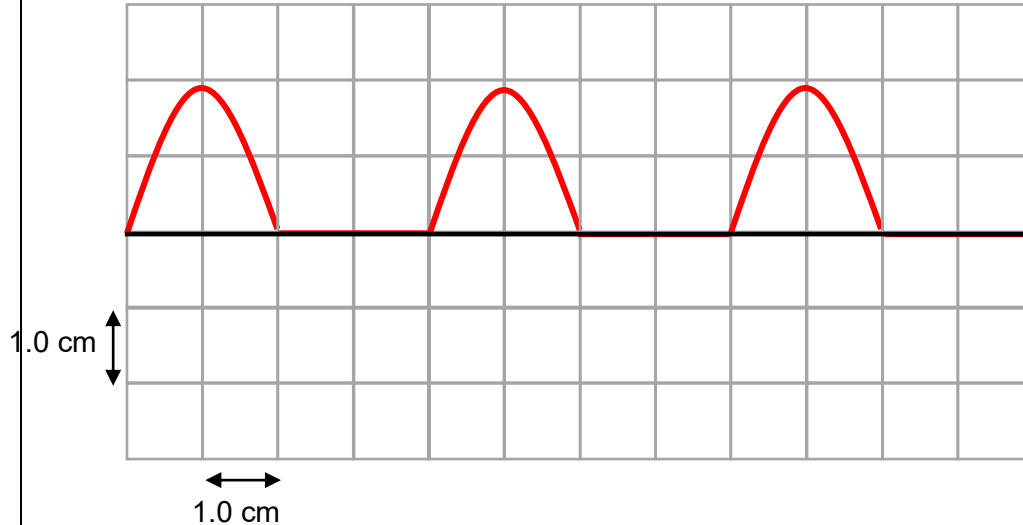
iii
(1)

Shape (half wave) – 3 complete cycles

Correct period & amplitude

$$T = 1/25 = 40 \text{ ms}$$

$$V_o = 9.9 \text{ V}$$



iii(2)

$$P_{\text{mean,rectified}} = \frac{P_o}{4} = \frac{V_o^2}{4R}$$

$$R = \frac{(\sqrt{2} \times 7)^2}{4 \times 4.5}$$

$$= 5.44 = 5.4 \Omega$$